Fiscal Year:	FY 2009	**Task Last Updated:**	FY 11/05/2008
PI Name:	Soller, Babs R. Ph.D.		
Project Title:	Noninvasive Biosensor Algorithms for Continuous Metabolic Rate Determination		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Westboro	**State:**	MA
Zip Code:	01581	**Congressional District:**	3
Comments:	Previous affiliation was University of Massachusetts Medical School, Worcester, MA, until October 2010 (Ed., 11-9-2010)		
Project Type:	Ground	**Solicitation / Funding Source:**	2007 NSBRI-RFA-07-01 Human Health in Space
Start Date:	10/01/2007	**End Date:**	09/30/2010
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: PI changed affiliation around October 2010 and new grant number issued, changing this end date to 9/30/2010, per NSBRI (11/9/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lee, Stuart (Wyle Integrated Sciences and Engineering Group)		
Grant/Contract No.:	NCC 9-58-SMS01301		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Vision for Space Exploration calls for a return to the moon in the next decade; NASA is planning for extended human presence on the lunar surface with both scientific and commercial objectives. Significant astronaut activity will be conducted in a spacesuit or EVA (extravehicular activity) suit. The suit becomes the astronaut's personal habitat, supplying O2, removing CO2, and maintaining appropriate temperature. During lunar surface exploration, the astronauts must have continuous real-time feedback to assure that all consumables (O2, cooling water, power) last until they can return safely to their quarters. The consumption rate of these resources depends upon the metabolic rate of the astronaut during EVA tasks and the suit's ability to manage thermal loads. This project will develop novel algorithms for our existing near infrared spectroscopy (NIRS) platform for real-time assessment of metabolic rate (measured as the rate of oxygen consumption, VO2) and muscle temperature. This capability is intended to be incorporated into biosensors which will be part of a smart system to advise astronauts about their usage of consumables during lunar surface activities. The following specific aims were proposed: 1. Develop and validate algorithms to accurately calculate VO2 from NIR spectra collected from muscle; 2. Develop and validate algorithms to simultaneously calculate muscle temperature; 3. Support incorporation of the sensor algorithms into the EVA suit testing program, where practical. We have demonstrated the feasibility determining VO2 with the Fick equation and data that can be collected solely through a NIRS system. The accuracy of this technique is currently limited at high exercise intensities, primarily due to inaccuracies in our stroke volume estimate. We have initiated human studies that will allow us to improve our ability to estimate stroke volume from heart rate, increase the accuracy of our VO2 calculation and demonstrate we can detect a decrease in VO2 as a result of deconditioning (bed rest). Over the next year we expect to complete bed rest on 10 subjects. We also have been working on improving the accuracy of the NIRS measurements used to calculate oxygen content. These improvements are expected to be complete over the next year and will be tested with the data collected from bed rest subjects as well as data collected from a hypovolemia study, that will look at the effect of plasma volume changes alone on these measurements. We have entered into a new collaboration, supported by non-NASA funds, to develop a solid-state, low profile spectroscopic sensor which will have the potential to provide a prototype unit with appropriate characteristics to be used for ground testing within the EVA suit. Non-suit testing of this sensor at JSC is expected begin this calendar year. The proposed biosensor system has built-in redundancy, through easy application to both limbs, and will provide system redundancy in the measurement of heart rate and temperature to further assure astronaut safely. The proposed technology is expected to result in wearable sensors that terrestrial doctors and their patients can use to track and optimize exercise in the management health and fitness, as well as during related applications in the care of critically ill patients.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will have direct earth based application. The fitness and exercise applications we are developing can be used to assist in the training and evaluation of elite and recreational athletes. This direct muscle application of interest to NASA for assessing fitness in space may be useful to assess success of physical therapy in rehabilitating patients with muscle injury or atrophy. The sensor, which also is of tremendous interest to the Army, will have application in emergency response vehicles, emergency rooms, and hospitals. Pre-hospital applications include assessing the severity of shock and triaging multiple casualties, as well as providing a sensor for a smart medical system to guide resuscitation from hemorrhage. In the ICU we expect that this monitor will find application in helping provide early identification of patients with hemodynamic instability before they go into shock. The miniaturization of the sensor and monitor, required for EVA suit placement, will result in a highly portable system for emergency medical use. If small and inexpensive enough, it could be used world-wide for screening of anemia associated with malnutrition.
Task Progress:	We developed a method to estimate stroke volume from heart rate and used it to calculate VO2 from the Fick equation using NIR spectra to calculate oxygen content. This was demonstrated during an incrementally increasing cycle exercise. We found very good agreement between VO2 measured by standard ventilatory methods and NIRS-determined VO2 at exercise levels below the anaerobic threshold. Significant deviations were observed at higher exercise intensities. In the first year of this project our task was to improve the accuracy of our VO2 calculation. The two components of this are an improvement in the spectroscopic calculation of hematocrit (Hct) and venous oxygen saturation (SvO2), as well as an improved method for estimating stroke volume. We presented the results of our initial VO2 calculations to physiologists at the ACSM and AsMA meetings. Feedback we received indicated that the largest source of error was most likely in our estimate of stroke volume at high exercise intensities. Over the last year we have developed a new calibration methodology that we expect will improve the accuracy of Hct and SvO2 measurement. This methodology is currently being tested in phantom materials and with previously collected data. To improve our method of estimating stroke volume we have worked with the JSC Cardiovascular Laboratory to develop a method to determine stroke volume at each stage of the standard VO2max protocol. This technique was developed and tested on four pilot test subjects where we simultaneously measured VO2 and stroke volume with echocardiography. We received approval to add NIRS and echocardiography to the current bed rest campaign at UTMB and we have subsequently collected NIRS, echo and VO2 data from 5 bed rest subjects in their R-12 and R-7 testing. These subjects will be measured again at the end of their 90 day bed rest period. We also have been approved to add these additional measures to the cycle testing for the next planned bedrest campaign to imrpove the rate of our data collection. We have received synergistic funding from non-NASA sources to develop a solid state, low profile sensor that can be worn in a space suit for determining metabolic rate.
Bibliography Type:	Description: (Last Updated: 03/22/2019)

Abstracts for Journals and Proceedings	Soller BR, Yang Y, Lee SM C, Wilson CA. "Comparison of noninvasively determined hydrogen ion and lactate thresholds during cycle exercise." American College of Sports Medicine 55th Annual Meeting, Indianapolis, IN, May 28-31, 2008 Med Sci Sports Exerc. 2008;40(5 Suppl):S426. http://dx.doi.org/10.1249/01.mss.0000322816.61090.71 , then click LWW access. , May-2008
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Articles in Peer-reviewed Journals	Soller BR, Yang Y, Lee SM, Wilson C, Hagan RD. "Noninvasive determination of exercise-induced hydrogen ion threshold through direct optical measurement." J Appl Physiol. 2008 Mar;104(3):837-44. PMID: 18096753 , Mar-2008
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Papers from Meeting Proceedings	Marengi N, Yang Y, Lee SM C, Wilson C, Heard SO, Soller BR. "Feasibility of using near infared spectroscopy in determining VO2 for preoperative risk assessment." American Society of Anesthesiologists Annual Meeting, San Francisco, CA, October 13-17, 2007. Foundation for Anesthesia Education and Research (FAER) Symposium, Americal Society of Anesthesiologists Annual Meeting, October 2007. , Oct-2007
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Patents	Patent Application 12/172,942. Patent July 2008. Jul-2008 Soller BR. "Physical Performance Monitoring and Monitors."